

THOMSON TRI-TOWNSHIP, IL, USA

WMO: 722204

Lat: 42.046N

Lon: 90.108W

Elev: 188

StdP: 99.09

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04996

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-17.9	-25.9	0.4	-20.5	-22.7	0.5	-17.5	10.6	-4.8	9.2	-3.4	1.8	290	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.9	23.3	31.6	22.9	30.1	22.2	25.3	30.7	24.4	29.5	23.5	28.2	4.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.8	19.1	28.2	22.8	17.9	27.2	22.2	17.2	26.8	78.5	31.1	74.6	29.3	71.1	28.2	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
9.3	8.3	7.4		-27.4	35.3	4.9	1.9	-30.9	36.6	-33.8	37.7	-36.6	38.8	-40.1		
				-27.5	27.1	4.7	1.0	-30.9	27.8	-33.6	28.3	-36.2	28.9	-39.6	40.1	29.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-5.5	-4.6	3.8	10.3	16.6	21.6	23.1	21.8	18.7	11.3	4.2	-2.6
	DBStd	11.43	6.94	6.39	6.37	5.13	4.96	3.25	3.26	3.02	4.30	5.19	5.35	6.04
	HDD10.0	1788	482	408	211	58	6	0	0	0	1	46	186	391
	HDD18.3	3534	740	641	452	244	94	8	2	6	47	227	424	649
	CDD10.0	1777	0	0	19	67	211	348	407	366	262	86	11	1
	CDD18.3	483	0	0	2	4	41	106	151	113	58	8	0	0
	CDH23.3	4543	0	0	14	63	399	964	1491	992	539	81	0	0
	CDH26.7	1514	0	0	2	11	120	318	565	311	171	17	0	0
Wind	WSAvg	3.2	3.4	3.4	3.7	4.2	3.5	2.8	2.3	2.2	2.6	3.1	3.5	3.3
Precipitation	PrecAvg	922	36	39	57	89	108	130	94	108	90	65	65	40
	PrecMax	1208	81	92	131	169	218	264	180	190	162	129	159	107
	PrecMin	684	10	4	14	35	18	37	18	28	24	12	1	7
	PrecStd	122	18	25	41	38	49	62	40	48	39	38	44	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.5	25.4	27.9	32.7	33.9	36.3	33.7	33.1	28.8	21.3	14.8
		MCWB	7.6	8.5	17.5	16.0	21.6	23.6	25.3	22.8	23.5	20.4	14.7	12.6
	2%	DB	7.5	7.9	19.9	25.2	29.8	32.2	33.0	31.4	31.1	25.8	17.5	10.0
		MCWB	5.6	5.2	14.1	15.6	19.6	22.6	24.1	23.3	22.3	18.5	12.5	8.2
	5%	DB	5.0	5.2	17.0	22.3	27.6	30.1	31.6	29.9	28.5	22.6	14.9	7.4
		MCWB	3.2	2.8	11.8	13.8	18.4	21.9	23.4	22.5	20.6	16.7	11.1	5.8
	10%	DB	2.8	3.0	13.0	19.8	25.5	28.0	29.9	28.0	26.9	20.0	12.3	5.0
		MCWB	1.7	1.6	8.8	12.7	17.8	20.8	23.0	21.3	19.7	15.6	9.1	3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	8.8	17.6	18.5	23.2	25.9	27.3	25.7	24.7	22.3	16.0	13.2
		MCDB	10.0	11.0	23.6	23.8	30.4	31.8	32.2	30.4	30.4	27.6	19.1	14.3
	2%	WB	5.3	5.5	15.2	16.7	21.7	24.2	25.7	24.7	23.5	20.1	13.6	8.3
		MCDB	7.3	8.1	19.3	22.3	27.4	29.7	31.2	29.2	28.4	23.5	16.7	9.3
	5%	WB	3.4	3.2	12.4	15.2	20.3	23.3	24.9	23.6	22.3	18.2	11.6	5.8
		MCDB	4.8	4.8	15.6	21.1	25.4	28.3	30.1	28.0	26.8	21.1	14.0	7.0
	10%	WB	1.8	1.9	9.2	13.6	18.9	22.2	23.9	22.6	21.0	15.9	9.3	3.5
		MCDB	3.0	3.1	13.0	18.9	23.6	26.9	28.4	26.7	24.7	19.5	12.1	4.9
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.9	9.8	11.5	11.9	11.3	11.4	11.4	12.6	11.1	9.2	7.6
		MCDBR	10.0	10.1	14.4	16.2	14.3	13.2	12.8	12.8	14.1	14.6	13.0	10.1
		MCWBR	8.0	7.8	8.6	8.5	6.7	6.1	5.5	5.8	6.3	8.0	8.7	8.3
	5% WB	MCDBR	9.5	9.8	12.5	13.5	12.3	11.5	11.2	10.5	11.5	11.6	11.8	9.6
		MCWBR	7.9	7.9	8.4	8.3	6.4	6.1	5.8	5.5	5.9	7.7	8.7	8.3
Clear-Sky Solar Irradiance	taub	0.287	0.302	0.341	0.387	0.414	0.427	0.445	0.422	0.393	0.343	0.301	0.286	
	taud	2.401	2.372	2.372	2.257	2.244	2.253	2.193	2.256	2.313	2.450	2.501	2.429	
	Ebn at Noon	863	904	905	881	863	850	829	837	837	846	844	832	
	Edh at Noon	79	95	108	131	136	135	142	130	114	88	71	71	
All-Sky Solar Radiation	RadAvg	1.75	2.74	3.73	4.64	5.45	6.12	6.09	5.37	4.49	2.94	1.93	1.42	
	RadStd	0.21	0.31	0.35	0.39	0.38	0.49	0.38	0.35	0.37	0.35	0.16	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page